ODD+D Appendix for Comparing Legislative Mechanism Bundles

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This appendix documents the congressional institutional simulator using the ODD and ODD+D model-description conventions [Grimm et al. 2010; Müller et al. 2013]. It describes entities, state variables, scheduling, adaptation, input data, submodels, outputs, empirical flow sanity checks, assumptions, and reproducibility commands for the simulator used in *A Reproducible Simulation Framework for Comparing Legislative Mechanism Bundles*.

CCS Concepts: • **Computing methodologies** → **Modeling and simulation**; • **Human-centered computing** → *Collaborative and social computing theory, concepts and paradigms.*

Additional Key Words and Phrases: ODD+D, legislative simulation, institutional design, model documentation

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1 Appendix Roadmap

Table 1. Where to look for common reviewer questions.

Reviewer concern	Where to look
What assumptions generate the synthetic worlds?	Table 2 and Section 6
What is empirically checked?	Table 3 and Section 7
What weights enter the vote kernel?	Table 4
What are the selected main-campaign results?	Table 5; full CSV in `reports/simulation-campaign-v21-paper.csv`
What robustness probes exist?	Tables 6, 7, 8, and 9
What are the main limitations?	Section 11
How is the artifact reproduced?	Section 12

2 Purpose

The model compares legislative institutional designs under shared generated political worlds. Its purpose is to stress-test mechanisms that alter productivity, revision moderation, representative responsiveness, public-benefit alignment, minority protection, and resistance to capture. It is not a forecast of a specific Congress. It is a mechanism-comparison simulator: the same generated legislators and bills are routed through many institutional processes so differences can be attributed to procedural rules and incentive structures.

3 Entities, State Variables, and Scales

3.1 Legislators

Legislators have an identifier, party label, ideology, revision-moderation preference, party loyalty, constituency sensitivity, lobbying susceptibility, reputation sensitivity, district preference, district intensity, affected-group sensitivity, and a representation profile. The representation profile records chamber eligibility, represented population share, district

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magnitude, region label, elected/appointed mode, lower- and upper-house term lengths, renewal cohort, renewal cycle, next renewal round, and selector body. These values determine how the legislator weighs ideology, party pressure, constituent/public support, lobbying pressure, revision-moderation incentives, reputation, and concentrated harm before casting a final yes-or-no vote.

3.2 Bills

Bills have an identifier, title, proposer, proposer ideology, original and current policy position, public support, generated public benefit, public-benefit uncertainty, lobby pressure, private gain, salience, issue domain, anti-lobbying-reform flag, affected group, affected-group support, concentrated harm, compensation cost, cosponsor counts, amendment movement, lobbying channel spend, challenge status, citizen-review signals, active-law signals, and alternative-selection signals.

3.3 Lobby Groups

Lobby groups are explicit collective actors. Each group has an identifier, primary issue domain, issue-preference map, preferred policy position, budget, influence intensity, defensive multiplier, information bias, public-campaign skill, capture strategy, and public-support mismatch tolerance. Groups can spend through direct pressure, agenda-access pressure, information distortion, public campaigns, litigation or delay threats, and defensive anti-reform pressure.

3.4 Institutional Processes

An institutional process is a composable module. Processes can screen proposal access, apply leadership agenda control, modify public signals, spend lobbying or attention resources, route bills to procedural tracks, mediate amendments, evaluate affected-group harm, select among alternatives, conduct chamber votes, revise through conference committees, apply vetoes, apply judicial review, route through citizen initiatives or citizen agenda petitions, register enacted laws, or review laws later. The chamber-structure supplement adds chamber-specification, chamber-origin routing, second-chamber amendment and veto powers, last-offer bargaining, lower-house override, committee-assignment configuration, committee-role power, eligibility and retention filters, and ex ante review through insulated fiscal, electoral, audit, ethics, or legal-review bodies. The simulator also includes bounded linked modules for open floor calendars, discharge-petition committee bypass, district-population public signals, pairwise amendment tournaments, long-horizon proposer strategy, omnibus bargaining, campaign-finance influence systems, and constitutional-court architecture. The final decision still resolves to a yes-or-no outcome, enactment or rejection, and a post-bill status quo. Portfolio variants combine low-risk fast passage, middle-risk pairwise alternatives and mediation, high-risk citizen and harm review, proposal bonds, anti-capture safeguards, law-registry correction, and, in the expanded variant, public-opinion, influence-system, omnibus, and court-review proxies.

4 Process Overview and Scheduling

Each run follows a fixed schedule:

(1) Generate a world containing legislators, party positions, lobby groups, a status quo, and a bill stream.
(2) Build each scenario's legislative process from that world.
(3) For each bill, create a deterministic vote context containing party positions, status quo, and seeded randomness.
(4) Pass the bill through the scenario process chain.

(5) Record the bill outcome, including agenda disposition, vote totals, enactment, amendments, challenges, lobbying signals, public-review signals, and policy movement.
(6) Update the scenario's status quo when policy is enacted or rolled back.
(7) Aggregate outcomes into scenario metrics.

Campaigns repeat this over many runs and assumption cases. Scenarios share generated worlds but use independent deterministic random streams, supporting paired comparison among institutions.

5 Design Concepts

5.1 Basic Principles

The model assumes legislative outcomes are shaped by proposal access, agenda scarcity, voting thresholds, committee and information structures, lobbying and capture, amendment opportunities, public review, concentrated harm, reversibility, and strategic actor adaptation.

5.2 Emergence

Productivity, revision moderation, capture, volatility, legitimacy, and gridlock emerge from bill-by-bill interactions. No actor directly optimizes the reported global metrics.

5.3 Adaptation

Proposers adapt over repeated bills. They can reduce proposal volume, delay timing, moderate risky bills, seek outside-bloc cosponsorship, reduce lobby exposure, add compensation amendments, lower risk tolerance after bad outcomes, and withdraw weak risky bills. Proposer state includes trust, proposal pace, risk tolerance, concession rate, lobby exposure, cooldown, and streak counters.

Lobby groups adapt over repeated bills. They track returns by channel, select capture strategies from learned returns and current bill conditions, change issue-specific spending multipliers, adjust overall budget intensity, increase reform-threat sensitivity when anti-lobbying bills become threatening, and maintain defensive pressure against reforms that would reduce lobbying influence.

Institutional state also adapts through proposal credits, proposal bonds, law registries, sunset review, challenge-token exhaustion, attention-token spending, norm erosion, judicial reversals, and active-law correction.

5.4 Objectives

Legislators vote according to weighted incentives. Proposers seek agenda access and enactment while learning to avoid reputational or procedural failure. Lobby groups seek policy movement toward their issue preferences and private gains while resisting anti-lobbying reforms. Institutions impose constraints and incentives whose tradeoffs are measured afterward.

5.5 Learning and Prediction

Learning is deterministic and bounded rather than a full reinforcement-learning procedure. Actors use local outcome signals: enactment, public support, public benefit, capture risk, concentrated harm, salience, private gain, prior failures, and prior channel returns.

Table 2. Detailed generator variables and where they enter the model. This table expands the simplified assumptions table in the main paper.

Component	Variable	Distribution or formula summary	Scenario tuned?	Main use
Legislator ideal point	z_i	faction centers at $\{-s, 0, s\}$, $s = 0.25 + 0.65\rho$, Gaussian spread, clamp $[-1, 1]$	yes	vote kernel, parties
Party and district signals	party, d_i	party bins or party-system profile; $d_i = 0.78z_i + N(0, 0.18 + 0.10\rho)$	yes	party pressure, district diagnostics
Sensitivities	$c_i, \pi_i, \kappa_i, \lambda_i, r_i$	scenario center plus bounded Gaussian noise	yes	revision moderation, party, constituency, lobby, reputation weights
Status quo	q_0	$N(0, 0.18)$, clamp $[-1, 1]$; enacted bills update q_t	no	status-quo-relative voting
Bill position	x_b	proposer-centered $z_p + N(0, 0.22)$, with adversarial shock variants	yes	vote kernel, policy shift
Public support/benefit	p_b, g_b	$g_b = 0.34 + 0.34(1 - \lvert x_b \rvert) - 0.08 private + N(0, 0.18)$; $p_b = g_b + 0.08\ell_b - 0.05 private + N(0, 0.22)$	yes	support, yield, risk
Lobby and private gain	$\ell_b, private_b$	$\ell_b = N(0, L)$, $private_b = 0.72 \max(0, \ell_b) + U(0, 0.20)$, bounded	yes	access, capture, vote
Harm and uncertainty	h_b, σ_b	harm rises with distance, salience, private gain, and noise; uncertainty rises with salience, signal mismatch, and lobbying	yes	harm thresholds, risk metrics

5.6 Stochasticity

World generation, committee composition, citizen panels, bill signals, party-system profiles, and some routing decisions use seeded randomness. Identical seeds reproduce identical outputs.

5.7 Observation

The simulator reports productivity, floor load, enacted support, generated public benefit, cooperation, revision moderation, gridlock, access denial, committee rejection, challenge rate, chamber low-support passage, low-public-support enactment, policy shift, proposer gain, lobby capture, public alignment, anti-lobbying success, private-gain ratio, lobbying spend by channel, public benefit per lobby dollar, amendment rate, amendment movement, amendment overload, poison-pill rate, support gain after amendment, minority harm, concentrated-harm passage, compensation rate, legitimacy, enacted-policy quality, policy yield per submitted proposal, blockage reliance, public support, active-law welfare, implementation delay/capacity/failure risk, nonenforcement and underfunding risk, renewal pressure, reversal rate, correction delay, low-support active-law share, selected-alternative distance, proposer agenda advantage, alternative diversity, status-quo win rate, citizen-review metrics, citizen agenda petition metrics, open-calendar metrics, attention spending, objection-window use, public-will review, district alignment, cosponsorship, bond forfeiture, strategic decoys, proposer-access Gini, vetoes, overrides, chamber conflict, malapportionment, population-seat distortion, committee capture, eligibility exclusion, renewal staggering, recusal, ex ante review, review delay, and quiet-capture risk.

6 Initialization

`WorldGenerator` creates legislators from party-system and ideology parameters, derives party positions, generates lobby groups by issue domain, chooses a scalar status quo, and generates bills around proposer ideal points. Generated values are bounded by model validation. Party-system profiles include ideological bins, two major parties with minor parties, fragmented multiparty systems, dominant-party systems, and weighted campaign profiles.

7 Input Data and Empirical Flow Checks

Normal scenario runs use synthetic generated worlds. Empirical flow sanity checks read the tracked benchmark extract at `data/calibration/empirical-benchmarks.csv`. The extract maps empirical sources to simulator metrics and broad benchmark ranges. Named sources include Voteview roll-call data [Lewis et al. 2021], Congress.gov and govinfo bill histories [Library of Congress 2026; United States Government Publishing Office 2020], Comparative Agendas Project topic data [Comparative Agendas Project 2026], ParlGov party-system data [Döring and Manow 2024], U.S. Lobbying Disclosure Act filings [United States Senate Office of Public Records 2026], Center for Effective Lawmaking scores [Center for Effective Lawmaking 2026; Volden and Wiseman 2014], and V-Dem institutional indicators [V-Dem Institute 2026].

Table 3. Empirical boundary. Rows marked empirical check have shaped raw inputs and simulator proxy summaries; rows marked synthetic only are still modeled without raw-data support.

Area	Status	Claim boundary / next source
Roll-call behavior	empirical check (ready)	Party-unity and coalition-size proxies are inspectable. Next: Extend Voteview coverage across more Congresses and chambers.
Bill progression	empirical check (ready)	Floor load, committee reporting, and enactment-rate proxies are inspectable. Next: Replace the bounded sample with a full Congress.gov/govinfo bill census.
Lobbying disclosure	empirical check (ready)	Lobby-spend concentration is inspectable, but not yet tied to bill-level outcomes. Next: Link LDA clients/issues to bill topics and committee referrals.
Topic throughput	empirical check (ready)	Agenda diversity and topic enactment proxies are inspectable. Next: Map topics to CAP/congressional subject taxonomies across sessions.
Sponsor access	empirical check (ready)	Proposal-access concentration is inspectable with a rough sponsor sample. Next: Replace with CEL-style sponsor effectiveness or complete sponsor histories.
Public support	synthetic only (high)	District public-will and affected-group legitimacy remain synthetic. Next: Add district-level MRP/CCES-style opinion, turnout, and affected-population estimates.
Committee gatekeeping	empirical check (ready)	Reporting and referral proxies are inspectable; hearings remain rough. Next: Add committee hearing calendars, markup records, amendments, and discharge petitions.
Campaign finance	synthetic only (high)	Money-in-politics effects remain represented by lobbying proxies. Next: Add FEC/OpenFEC receipts, independent expenditures, and industry classifications.
Implementation feedback	synthetic only (high)	Implementation delay, nonenforcement, and capacity feedback remain synthetic. Next: Add Federal Register, Unified Agenda, Regulations.gov, and agency enforcement rows.
Law revision	synthetic only (high)	Law-registry correction and sunset dynamics remain synthetic. Next: Add statutory lineage for amendment, reauthorization, repeal, expiration, and invalidation.

The executable checker runs conventional scenarios and writes CSV and Markdown pass/fail reports under `reports/`. The pass is an empirical screen, not an automatic parameter fitter. Its benchmark ranges are intentionally broad and several are abstract proxies, so passing the screen should be read as minimum flow plausibility rather than validation. The empirical comparison checks include six documented samples: Congress.gov bill progression, Voteview roll calls, Senate LDA lobbying disclosures, and Congress.gov-derived topic, sponsor, and committee summaries. These samples support bill-attrition, floor-load, committee-report, coalition-size, party-unity, sponsor-concentration, topic-throughput, and lobbying-concentration checks. Optional API fixture samples are stored separately from raw empirical inputs; they test adapter shapes but are not used as validation evidence.

8 Submodels

8.1 Voting

Legislators cast `YAY` or `NAY` through weighted influences: ideology, party, constituency/public support, lobbying, revision-moderation preference, reputation, and affected-group sensitivity.

Table 4. Vote-score feature functions and weights. The anti-capture kernel changes weights; affected-group variables enter procedural thresholds and metrics rather than the base yes/no vote score.

Feature	Function summary	Standard w_k	Anti-capture w_k
Ideology	$(u_i(x_b) - u_i(q_t))(0.75 + salience_b)$	1.45	1.45
Party	party loyalty times $u_{party}(x_b) - u_{party}(q_t)$	0.85	0.78
Constituency/public	constituency sensitivity times $2(p_b - 0.5)$	0.95	1.12
Lobbying	lobby sensitivity times ℓ_b	0.55	0.14
Revision moderation	revision-moderation preference times 0.45 moderation +0.35 public legitimacy +0.20 incrementalism	0.75	0.82
Reputation	reputation sensitivity times unpopular-bill and lobby-capture penalties	0.55	0.92
Noise	seeded Gaussian perturbation before logistic voting	sd 0.25	sd 0.25

8.2 Proposal Access and Agenda Scarcity

Access rules include open access, open floor calendars with abusive-access screens, citizen agenda petitions, viability screens, scalar proposal costs, public-value waivers, lobby surcharges, member quotas, leadership agenda control, public-interest screens, anti-capture proposal-access screens, cross-bloc cosponsorship gates, affected-group sponsor gates, proposal bonds, earned proposal credits, agenda lotteries, and quadratic attention budgets.

8.3 Committees and Information

Committee gatekeeping can reject proposals before floor consideration. Committee information processes move public-signal estimates toward generated public benefit with composition-dependent accuracy. Configured committee selection can vary issue domain, random seed, party quota, proportional assignment, expertise weighting, independence weighting, leadership control, rotation memory, chair allocation, quorum party balance, citizen advisory weight, and topic-referral authority. Committee-power processes distinguish priority queues, veto players, fast-track certification, scrutiny and audit, public accounts, legal/drafting review, and discharge-petition targets.

8.4 Chamber Structure, Eligibility, and Independent Review

Chamber specifications define equal-population, proportional, mixed-member, territorial, malapportioned, appointed, indirect, functional, or mixed elected/appointed chambers. Bicameral routing variants can change origination, emergency routing, second-chamber preclearance, amendment-only powers, absolute or territorial vetoes, suspensive vetoes, navette, mediation, last-offer bargaining, joint sittings, and lower-house overrides. Eligibility and retention filters model stylized age, residency, language, expertise, conflict, attendance, capture-risk, dual-office, party/executive overlap, recusal, cooling-off, term-limit, recall, public-support retention, and sanctions rules. Ex ante review variants model optional referral, mandatory high-risk review, advisory opinions, suspensive objections, mandatory clearance, qualified override, and deadline-limited review by insulated fiscal, electoral, audit, ethics, cost-estimator, regulator, or boundary institutions.

8.5 Challenge and Objection

Challenge vouchers allocate scarce blocking rights to parties or members. Q-member escalation lets a sufficient number of legislators trigger active review. Public objection windows route contested bills to active voting, higher thresholds, or repeal review depending on scenario.

8.6 Lobbying

Budgeted lobbying modifies bill pressure and public signals through channel-specific spending. Anti-capture variants include transparency, public financing, public advocates, blind review, audits, sanctions, and defensive-spend caps. Adaptive lobbying learns from channel returns and reform threats.

8.7 Amendment and Mediation

Structured mediation moves risky proposals toward weighted institutional targets. Multi-round mediation models repeated amendments, costs, concession limits, poison-pill risk, capacity pressure, public-support improvement against a no-amendment counterfactual, proposer-advantage reduction, and compensation for concentrated harm.

8.8 Distributional Harm

Harm-weighted thresholds, compensation amendments, and affected-group consent mechanisms distinguish broad aggregate benefit from concentrated minority loss.

8.9 Citizen Mini-Publics

Citizen panels simulate sampling noise, information quality, manipulation risk, informed support, benefit estimates, and legitimacy. Panel output can affect burden-shifting eligibility, active-vote routing, final threshold, agenda priority, or affirmative majority burden of proof. Citizen agenda-petition paths separate raw support from costly participation signals, public-support intensity, and cheap signal distortion caused by organized amplification.

8.10 Leadership, Direct Democracy, and Judicial Review

Leadership agenda control schedules bills using observable public support, salience, uncertainty, majority-party fit, capture risk, and concentrated-harm risk rather than generated true welfare. Citizen-initiative processes let sufficiently salient, high-support proposals bypass legislative floor blockage through a public referendum. Conference-committee processes revise near-miss bicameral bills toward a bargaining target before chamber revotes. Judicial-review processes can invalidate enacted laws with high rights-like harm, uncertainty, policy shock, or capture risk.

8.11 Alternatives, Package Bargaining, Norm Erosion, and Law Registry

Competing-alternative processes generate same-domain alternatives and select by public benefit, public support, chamber-median proximity, or pairwise majority before final ratification. Pairwise amendment tournaments use the same selector logic for amended versions, making amendment control an explicit pre-ratification stage rather than only a final yes/no vote. Package-bargaining processes approximate revision through side-payment costs, implementation-delay costs, modest policy movement, and harm-reducing package trades. Norm-erosion processes raise procedural delay and reduce public support signals under high contention. Law-registry processes allow enacted laws to persist, face implementation delay, operate under capacity/nonenforcement/underfunding risk, be renewed under lobbying pressure, and be repealed or partially rolled back after review.

9 Experimental Design

The main submitted evidence comes from the main comparison campaign. That campaign joins broad assumption cases, adversarial proposal-generator cases, party-system sensitivity cases, and a stylized rising-contention timeline in

Table 5. Selected core metrics from the main campaign, grouped by mechanism family. Entries are median [IQR] across broad and adversarial cases; the appendix reports mean/range, yield, seed diagnostics, and separate PAIR/AMT rows.

Label	Representative system	Prod. ↑	Rev. mod. ↑	Public ↑	Low supp. ↓	Risk ctrl. ↑	Admin ↓	Example profile ↑
CUR*	**Stylized U.S.-like benchmark**	**0.05 [0.03,0.06]**	**0.37 [0.36,0.38]**	**0.79 [0.78,0.80]**	**0.01 [0.00,0.01]**	**0.94 [0.92,0.94]**	**0.02 [0.01,0.02]**	**0.57 [0.56,0.58]**
SM	Simple majority	0.27 [0.22,0.32]	0.23 [0.23,0.25]	0.69 [0.67,0.70]	0.19 [0.18,0.23]	0.87 [0.85,0.88]	0.07 [0.07,0.07]	0.54 [0.53,0.55]
COMM	Committee regular order	0.16 [0.13,0.20]	0.28 [0.27,0.29]	0.74 [0.73,0.74]	0.09 [0.08,0.10]	0.91 [0.88,0.92]	0.03 [0.03,0.03]	0.55 [0.54,0.56]
DIS	Discharge petition	0.15 [0.09,0.17]	0.31 [0.30,0.32]	0.75 [0.74,0.77]	0.03 [0.03,0.05]	0.93 [0.92,0.94]	0.08 [0.07,0.08]	0.56 [0.55,0.56]
OPEN	Open floor calendar	0.31 [0.25,0.38]	0.24 [0.23,0.26]	0.67 [0.67,0.68]	0.25 [0.22,0.26]	0.85 [0.84,0.86]	0.10 [0.09,0.10]	0.54 [0.53,0.56]
SEL	Content-selection family	0.54 [0.49,0.56]	0.50 [0.48,0.52]	0.79 [0.77,0.79]	0.00 [0.00,0.00]	0.94 [0.93,0.95]	0.28 [0.28,0.29]	0.67 [0.66,0.69]
ACG	Anti-capture access	0.26 [0.20,0.31]	0.25 [0.23,0.26]	0.71 [0.68,0.72]	0.17 [0.15,0.20]	0.89 [0.88,0.90]	0.06 [0.06,0.07]	0.55 [0.54,0.56]
HARM	Harm-weighted majority	0.25 [0.18,0.27]	0.24 [0.23,0.26]	0.70 [0.68,0.70]	0.19 [0.18,0.23]	0.88 [0.87,0.89]	0.07 [0.07,0.07]	0.54 [0.53,0.55]
DP	Burden-shifting stress case	0.84 [0.84,0.89]	0.11 [0.10,0.13]	0.56 [0.54,0.57]	0.50 [0.47,0.52]	0.62 [0.61,0.63]	0.07 [0.07,0.07]	0.55 [0.54,0.56]
DPM	Burden shift + mediation	0.81 [0.72,0.89]	0.22 [0.21,0.26]	0.61 [0.59,0.64]	0.41 [0.34,0.43]	0.73 [0.71,0.75]	0.21 [0.19,0.22]	0.58 [0.57,0.60]
PORT	Portfolio hybrid	0.52 [0.45,0.58]	0.40 [0.40,0.45]	0.76 [0.76,0.78]	0.03 [0.01,0.04]	0.93 [0.93,0.93]	0.23 [0.23,0.24]	0.64 [0.63,0.66]

Note: Example profile is one revision-moderation-weighted sensitivity check:
0.30 Rev. mod. + 0.20 Prod. + 0.20 Risk ctrl. + 0.15 Admin feas. + 0.15 Public. Admin feas. is the inverse of the administrative-cost index shown in the Admin column. Low supp. is low-public-support enactment. The asterisk marks the stylized U.S.-like conventional benchmark. Cell values carry the comparison; shading follows the column arrow as a secondary cue.

Table 6. Appendix generator-sensitivity and fairness-control details. These rows summarize robustness probes; they are not validation evidence.

Type	Check	Result	Interpretation
Generator	Extreme-beneficial reform	top PORT 0.72; SEL 0.69; DPM 0.58; DP low supp. 0.57	Tests whether high-distance proposals can carry high generated public benefit.
Generator	Revision dilution	top PORT 0.71; SEL 0.67; DPM 0.57; DP low supp. 0.67	Tests whether moderation can reduce generated public benefit.
Generator	Lobby information	top SEL 0.74; PORT 0.70; DPM 0.65; DP low supp. 0.28	Tests useful organized information rather than only capture pressure.
Generator	Public-opinion error	top PORT 0.68; SEL 0.68; DPM 0.62; DP low supp. 0.41	Tests noisy or systematically biased public-support signals.
Generator	Minority-rights harm	top SEL 0.66; PORT 0.64; DPM 0.63; DP low supp. 0.19	Tests majority support paired with concentrated rights-like harm.
Generator	Technocratic delay	top PORT 0.69; SEL 0.66; DPM 0.60; DP low supp. 0.75	Tests low initial support with delayed generated public benefit.
Fairness	Conventional revision	CUR P 0.05, M 0.36, R 0.93, A 0.02; PROC P 0.05, M 0.32, R 0.93, A 0.03	Gives the conventional family conference/review content improvement.
Fairness	Majority mediation	SM P 0.33, M 0.24, R 0.85, A 0.07; SMED P 0.40, M 0.22, R 0.80, A 0.19	Compares simple majority with a mediation content-improvement step.
Fairness	Committee amendment	COMM P 0.22, M 0.28, R 0.89, A 0.03; CAMD P 0.12, M 0.32, R 0.93, A 0.07	Compares committee gatekeeping with committee information/amendment power.

Note: P is productivity, M is revision moderation, R is risk control, and A is administrative cost. Fairness rows compare content-improvement variants; they do not impose a hard cost-constrained reoptimization.

one artifact. The main manuscript focuses on the stylized U.S.-like benchmark, simple majority, committee regular order, discharge petition, open floor calendar, a combined content-selection family, anti-capture access, harm-weighted majority, a small burden-shifting stress-test family, and a portfolio hybrid. The appendix tables report separate pairwise-alternative and amendment-tournament rows plus the broader implemented catalog, including leadership agenda control, chamber and committee variants, citizen petitions and initiatives, mini-publics, agenda scarcity, package bargaining, review and correction systems, adaptive routing, norm erosion, and influence-system safeguards. Adversarial generator cases test high-benefit extreme reform, popular harmful bills, moderate capture, delayed-benefit reform, concentrated rights-like harm, anti-lobbying backlash, revision dilution, useful lobbying information, and public-opinion error. Follow-up diagnostics include mechanism ablations, manipulation-stress campaigns, chamber-structure family/stress screens, and empirical flow sanity checks that map optional raw empirical summaries to simulator proxy metrics.

The dense full scenario-average table is not repeated in the PDF appendix. Full numeric outputs, including separate pairwise-alternative and amendment-tournament rows, are provided in `reports/simulation-campaign-v21-paper.csv`; the generated LaTeX full table remains in `paper/figures/scenario_averages_full_table.tex` for local inspection.

The following sensitivity and fairness probes are not validation evidence. They test whether the framework exposes dependence on generator assumptions and whether major qualitative patterns survive limited alternative worlds.

10 Chamber-Structure Supplement

The chamber-structure campaign tests representation architecture, committee power, eligibility filters, and ex ante review separately from the main mechanism comparison.

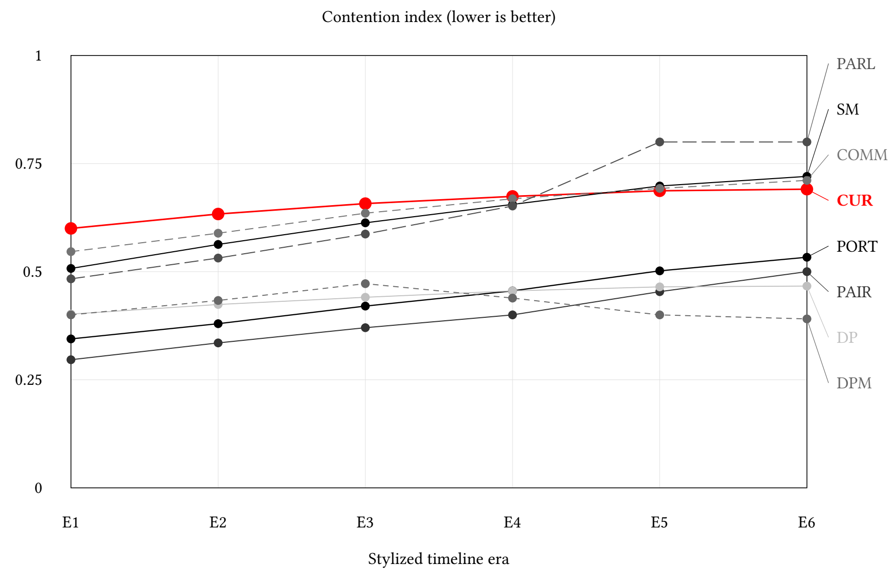

Fig. 1. Appendix rising-contention stress paths from the main comparison campaign. The index is 0.50·gridlock +0.30 · (1−revision moderation) + 0.20·low-public-support enactment. Lower is better.

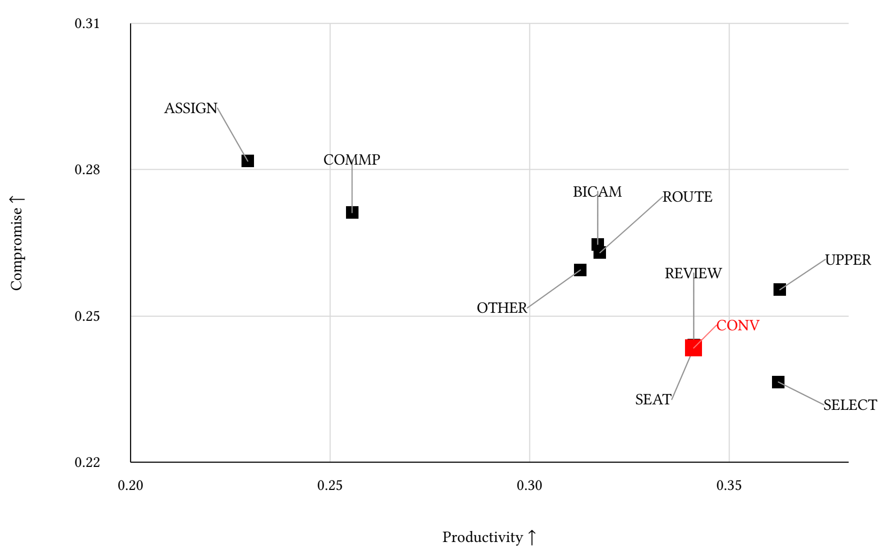

Fig. 2. Chamber-structure family champions from the supplemental chamber campaign, zoomed to the observed champion range so clustered family labels remain readable. The axes keep the paper's central productivity/revision-moderation comparison visible; malapportionment remains in Table 11 because the champion set has little variation on that metric.

Table 7. Objective-set frontier sensitivity.

Objective set	Frontier members	Interpretation
Productivity + revision moderation + risk	DP, DPM, SEL	Original narrow frontier.
Productivity + revision moderation + risk + admin	DP, DPM, SEL, PORT, OPEN, SM, HARM, ACG, COMM, DIS, CUR	Adding administrative feasibility makes most selected systems non-dominated.
Yield + risk + admin	DP, DPM, SEL, PORT, OPEN, SM, ACG, COMM, DIS, CUR	Prioritizes generated public benefit per submitted proposal.
Public support + harm + admin	SEL, PORT, DIS, CUR	Prioritizes support, affected-group protection, and feasibility.
Low-support avoidance + productivity	DP, DPM, SEL	Makes low-public-support avoidance a hard objective.

Note: This table is a sensitivity diagnostic. Frontier membership depends strongly on the objective set.

Table 8. Ablation diagnostics for selected mechanisms.

Mechanism	Removed or replaced component	Directional	Rev. mod.	Low supp. reduction	Admin added
Policy tournament	No alternatives	+0.056	+0.266	+0.217	+0.219
Risk routing	No jury lane	-0.002	+0.000	+0.001	+0.008
Anti-capture	Screen only	+0.024	+0.000	+0.043	+0.005
Package bargaining	Scalar package	-0.015	-0.002	+0.004	+0.065
Reversibility	No registry	-0.009	-0.009	-0.034	+0.044

Note: Positive values indicate how much the original mechanism improved relative to the ablated variant, except Admin added, where positive means additional procedural cost in the original mechanism.

Table 9. Manipulation-stress losses for selected mechanisms.

Stress case	Attack or perturbation	Directional loss	Rev. mod. loss	Low supp. added	Admin added
Tournament attack	Clones/decoys	+0.087	+0.046	+0.010	-0.077
Citizen-panel manipulation	Noisy panel	-0.002	+0.018	-0.005	+0.000
Bad-faith harm claims	Loose claims	-0.002	-0.018	-0.018	+0.000
Objection astroturf	Astroturf/noise	+0.012	-0.006	+0.003	+0.010
Agenda flooding	Proposal flood	+0.008	-0.011	+0.007	-0.001

Note: Positive values are losses or added burden under bounded manipulation stress. Low supp. is low-public-support enactment.

Table 10. Representative-versus-family-champion audit. Scores are from the fixed supplemental family-screen baseline, not the main comparison campaign. Negative gaps show cases where the paper uses a readable representative rather than the within-family top scorer.

Label	Family	Family-screen champion	Rep. dir.	Champ. dir.	Gap
DIS	Agenda gate	Committee-first regular order	0.626	0.637	-0.010
OPEN	Other	Long-horizon strategic learning	0.635	0.665	-0.030
DPM	Default-pass side family	Default pass unless 2/3 block	0.689	0.723	-0.035

11 Assumptions and Limitations

The policy space is one-dimensional. Public benefit is generated rather than empirically estimated. Legislators are synthetic and do not represent named real officials. Lobby groups are abstract collective actors. The empirical flow

Table 11. Chamber, apportionment, committee, and review family champions from the chamber-structure supplement. Values are averaged across the chamber campaign cases.

Label	Family champion	Prod.	Comp.	Risk	Malapp.
BICAM	Bicameral conflict	0.32	0.26	0.86	0.00
ASSIGN	Committee assignment	0.23	0.28	0.89	0.00
COMMP	Committee power	0.26	0.27	0.87	0.00
CONV	Conventional benchmark	0.34	0.24	0.84	0.00
REVIEW	Independent review	0.34	0.24	0.84	0.00
ROUTE	Origination and routing	0.32	0.26	0.85	0.00
OTHER	Other chamber design	0.31	0.26	0.85	0.00
SEAT	Seat allocation	0.34	0.24	0.84	0.00
SELECT	Selection and retention	0.36	0.24	0.83	0.00
UPPER	Upper-chamber composition	0.36	0.26	0.84	0.12

layer screens plausible ranges but does not fit raw datasets automatically. Leadership, initiative, conference, judicial, court-architecture, district-public, influence-system, long-horizon learning, and norm-erosion processes are stylized mechanism prototypes rather than detailed institutional models. Institutional mechanisms are simplified so they can be compared in bundles without modeling the full legal, administrative, judicial, media, and electoral environment. Omnibus and package bargaining are proxies for side payments, delays, exemptions, jurisdictional trades, distributive load, and harm reduction, not full models of legal text, appropriations, or multidimensional coalition formation.

12 Reproducibility

The main reproducibility entrypoints are:

- `make test`: compile and run the simulator test suite.
- `make campaign` or `make paper-campaign`: regenerate the comparison campaign.
- `make calibrate`: run empirical flow benchmark screening.
- `make seed-robustness-check`: run multi-seed checks for paper scenarios.
- `make validation-readiness`: check optional raw-input shape.
- `make empirical-validation`: summarize any present raw inputs.
- `make empirical-bridge`: map raw summaries to simulator proxies for empirical flow sanity checks.
- `make validation-gap-report`: generate the paper-facing empirical-boundary report. Adapter fixtures live separately and do not count as raw empirical data.
- `make build-bill-progression-raw`: rebuild the documented Congress.gov bill-progression raw sample when a Congress.gov API key is available.
- `make build-core-raw-validation`: rebuild the broader Voteview, Congress.gov-derived, and Senate LDA empirical comparison samples when network access and any required API key are available.
- `make ablation-analysis`, `make manipulation-stress`, and `make mechanism-diagnostics`: run mechanism ablations, manipulation stress tests, and the generated diagnostics table workflow.
- `make family-screen`, `make catalog-breadth`, and `make chamber-structure`: run breadth-catalog and chamber-structure diagnostics.
- `make paper`: build the paper and appendix PDFs.
- `make reproduce-paper-offline`: run the no-network paper reproduction path from fixed seeds and local summaries.
- `make supplement-anonymous`: build the double-blind supplement package.
- `make public-provenance`: after review, generate a non-anonymous provenance manifest.

The review package includes generated campaign reports and PDFs. Public repository identifying details are withheld during double-blind review and can be restored after review if permitted.

Use of Large Language Model Tools

The research question and core simulator idea originated with the human author. OpenAI ChatGPT/Codex tools available in the authoring environment were used for grammar and style editing, restructuring suggestions, code debugging assistance, and drafting assistance for portions of the simulator implementation, generated-report scripts, and manuscript prose. Tool outputs were edited and checked by the author; references, code outputs, experiments, tables, and claims were manually reviewed against the repository artifacts before inclusion. No large language model is listed as an author.

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